Display Data: Histograms

Lesson 8-6

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

Bars side by side: 0-9 pts (3 players), 10-19 pts (8 players), 20-29 pts (4 players)

Histogram

A bar graph that groups data into equal ranges. The bars touch.

If 5 players scored 10-19 points, the frequency for that interval is 5

Frequency

How many times a value shows up.

0-9, 10-19, 20-29 are intervals of width 10 — each covers 10 values

Interval

A range of numbers used to group data.

Most data in the middle with fewer at the ends = bell-shaped; most on one side with a tail = skewed

Distribution

How the data is spread out.

A histogram that is tallest in the middle and shorter on both sides is symmetric

Data distribution

How the data looks: where it sits and how spread out it is.

Data in just 2 intervals = low variability. Data across 6 intervals = high variability

Variability

How spread out the numbers are.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can make and read a histogram to display data in intervals.

1️⃣ Notice

The basketball league has 30 players and wants to display everyone's points-per-game average in a way that shows how the data is distributed.

2️⃣ Key idea

I can make and read a histogram to display data in intervals.

3️⃣ Words I'll use

Histogram **Frequency** **Interval** **Distribution** **Data distribution** **Variability**

🤔 Think About It

- Where do most of the scoring averages seem to cluster?
- Are there many players at the very low or very high end?
- How could you group these numbers to see a pattern?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: A histogram shows these frequencies: 0–4: 3 players, 5–9: 8 players, 10–14: 12 players, 15–19: 5 players. How many players are represented in total?

A. 28

B. 12

C. 20

D. 15

Step 1 Add all frequencies: 3 + 8 + 12 + 5 = 28 players.

✅ **Answer:** A. 28

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: In a histogram, what does the height of each bar represent?

A. The frequency (count) of data in that interval

B. The average of the data in that interval

C. The range of the interval

D. The median of the data

Step 1 ______

Step 2 ______

Answer: ______

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: How is a histogram different from a regular bar graph?

A. Histogram bars touch (no gaps) because the intervals are continuous ranges

B. Histograms use colors and bar graphs don't

C. Bar graphs show numbers and histograms show words

D. There is no difference

Show your work:

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

You have 30 players' points-per-game averages. Why is it easier to display them in a histogram with intervals than to list all 30 numbers?

Level 1 support

Start here: A histogram groups data into equal intervals so large data sets become readable.

💡 **Need a hint?**

Sentence starters (optional)

A histogram is easier because it groups the data into ____.

Un histograma es más fácil porque agrupa los datos en ____.

Each bar shows how many players fall in that ____.

Cada barra muestra cuántos jugadores caen en ese ____.

Word bank: histogram | frequency | interval | distribution | bar

Listen for: *Students explain grouping into intervals makes 30 values readable and that each bar's height is the frequency for that interval.*

Level 2

Predict: how would the histogram change if you used intervals of 5 instead of 10? Justify the trade-off.

- With intervals of 5, there would be ____ bars and ____ detail.
- Smaller intervals show ____ but can make the shape look ____.

EXPLORE

As you build the frequency table with intervals of 10, what does the height of each bar in the histogram represent?

Level 1 support

Start here: The height of a histogram bar is the frequency of values in that interval.

💡 **Need a hint?**

Sentence starters (optional)

The height of each bar shows the ___ of that interval.

La altura de cada barra muestra la ___ de ese intervalo.

The tallest bar is the interval where ___.

La barra más alta es el intervalo donde ___.

Word bank: histogram · frequency · interval · distribution · bar

Listen for: *A strong answer states the bar height equals how many players' averages fall in that interval (the frequency) and locates the tallest bar.*

Level 2

Why can't you tell a single player's exact average just from the histogram? Justify using the intervals.

- I can't find one player's exact value because the bar only shows ___.
- The histogram groups values, so I lose ___.

CONNECT

A histogram of shots-per-game shows 0-2: 1 player, 3-5: 8, 6-8: 10, 9-11: 3, 12-14: 1. The coach says 'most players take a good number of shots.' Does the shape support that claim?

Level 1 support

Start here: The shape of a histogram shows where most of the data is concentrated.

💡 **Need a hint?**

Sentence starters (optional)

The histogram ___ the coach's claim because most players are in the ___ intervals.

El histograma ___ la afirmación porque la mayoría está en los intervalos ___.

The shape shows the data clusters around ___ shots.

La forma muestra que los datos se agrupan alrededor de ___ tiros.

Word bank: histogram | frequency | interval | distribution | shape

Listen for: *Students read the 3-5 and 6-8 intervals (18 of 23 players) as the cluster and judge whether 'a good number' fairly describes the shape.*

Level 2

Critique: a teammate ignores the 0-2 and 12-14 bars because they are short. Should those intervals be ignored? Defend.

- The short bars ___ be ignored because they still show ___.
- Even one player in the 12-14 interval tells us ___.

REFLECT

A histogram of heights shows 60-63 in: 2, 64-67 in: 7, 68-71 in: 9, 72-75 in: 4. Which interval has the most players, and how do you know?

Level 1 support

Start here: The tallest bar marks the interval with the greatest frequency.

💡 **Need a hint?**

Sentence starters (optional)

The ___ interval has the most players because its bar is ___.

El intervalo ___ tiene más jugadores porque su barra es ___.

I know because its frequency is ___.

Lo sé porque su frecuencia es ___.

Word bank: **histogram** **frequency** **interval** **distribution** **bar**

Listen for: *Students identify 68-71 in (9 players) as the most-populated interval because it has the highest frequency.*

Level 2

Generalize: what is one thing a histogram shows well, and one thing it cannot show, compared to a list of the raw data?

- A histogram shows ___ well because ___.
- It cannot show ___ because the data is grouped into ___.

Write About the Math

The Writing Revolution

I can explain my histogram using the words histogram, frequency, interval, and distribution.

1. Kernel Sentence

subject + verb

Model: Histogram is a bar graph that groups data into equal ranges. The bars touch.
Histograma es una gráfica de barras que agrupa datos en rangos iguales. Las barras se tocan.

Write a kernel sentence about histogram. Use a subject and a verb.
Escribe una oración base sobre histograma. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Histogram matters in math
Histograma importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Histogram matters in math because ____.
Histograma importa en matemáticas porque ____.

____

but
pero

Histogram matters in math, but ____.
Histograma importa en matemáticas, pero ____.

____

so
entonces

Histogram matters in math, so ____.
Histograma importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about histogram.
Di un hecho verdadero sobre histogram.

Histogram ____.

Question
Pregunta

Ask a question about histogram.
Haz una pregunta sobre histogram.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about histogram.
Muestra entusiasmo sobre histogram.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with histogram.
Dile a un compañero qué hacer con histogram.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

A histogram is easier because it groups the data into ____.
Un histograma es más fácil porque agrupa los datos en ____.

Each bar shows how many players fall in that ____.
Cada barra muestra cuántos jugadores caen en ese ____.

The height of each bar shows the ____ of that interval.
La altura de cada barra muestra la ____ de ese intervalo.

Try It

Solve on your own. Check the answer key when you are done.

1. Histogram bins are 0–9, 10–19, 20–29 with heights 4, 9, 2. How many data values total?

A. 15

B. 9

C. 6

D. 45

Show your work:

2. Why should histogram intervals be equal in width?

A. So the bars fairly compare how many values fall in each range

B. To make the graph colorful

C. Because data must be negative

D. So there are no numbers

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A teacher collected test scores and made two different histograms — one with intervals of 5 (50–54, 55–59, etc.) and one with intervals of 20 (50–69, 70–89, 90–109). Both use the same data. How might the two histograms look different? Which interval size gives you more detail about the distribution? When might the larger interval be better?

Sentence starter: The histogram with intervals of 5 would have ____ bars and show ____. The histogram with intervals of 20 would have ____ bars and show ____. Smaller intervals are better when ____, and larger intervals are better when ____.

Show your work:

Reflect — Exit Ticket

A histogram of player heights shows: 60–63 in: 2, 64–67 in: 7, 68–71 in: 9, 72–75 in: 4. Which interval has the most players?

A. 68–71 inches
B. 64–67 inches
C. 72–75 inches
D. 60–63 inches

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. The frequency (count) of data in that interval *— Each bar's height shows the frequency — how many data values fall within that interval.*
2. **You Try (notes):** A. Histogram bars touch (no gaps) because the intervals are continuous ranges *— In a histogram, bars touch because each interval connects to the next (0–9, 10–19, etc.) — the data is continuous.*
3. **Try It 1:** A. 15 *— 4 + 9 + 2 = 15 data values.*
4. **Try It 2:** A. So the bars fairly compare how many values fall in each range *— Equal widths make frequency comparisons fair across bins.*
5. **Exit Ticket:** A. 68–71 inches *— The 68–71 inch interval has the highest frequency of 9 players.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Histogram is a bar graph that groups data into equal ranges. The bars touch.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).